

Microservices Architecture – Design, Develop, Deploy

Assessment 2 – Programming Assignment 40%

INFS605 Microservices

Version 1.3

This assignment is worth 40% of your final grade for the course and is submitted on Canvas under the Assignments tab. It is due at **6PM on Friday 24 October 2025 New Zealand Daylight Time (NZST)**. This is an individual assignment.

Learning Outcome

Your learning outcome is to develop a microservices-based application with a focus on functional and non-functional requirements.

In this assignment you are to imagine that you are working for a small startup company called Osborne.AI that wants to prototype a lightweight microservices-based application. The product is to be a basic **Student Services Dashboard** which consists of a few lightly coupled services that simulate functionality relevant to university operations.

Your task is to design, implement, containerize, and deploy a prototype of a small **microservices-based system** using **Docker** and **Docker Compose** either a) that will run on **local hosting** (for example, running a local web server architecture from terminal on Visual Studio Code or a similar IDE) or b) on your **Ubuntu VM running on VirtualBox**. The goal is to demonstrate your understanding of microservices principles and container-based deployment.

You will work individually and submit your code via Canvas. Your system should be made up of a minimum of **3 new services**, running as separate Docker containers, and connected using docker-compose.

Technologies Required

- Ubuntu VM on VirtualBox (optional)
- Docker and Docker Compose
- Git or GitHub
- YAML
- PostgreSQL or MongoDB (you choose)
- (Optional) Python, Node.js, or Flask/FastAPI for backend services

Application Requirements

You will implement at least 3 new microservices from the list below (or propose your own):

Suggested Services:

- Student Profile Service – manages student data (name, email, programme, etc.).
- Course Catalogue Service – provides a list of courses with metadata.

- Feedback Service – allows students to submit feedback on courses.
- Notification Service – mock service that sends "emails" (log entries) when feedback is submitted.
- API Gateway (Optional Stretch) – routes requests to the correct microservices.

Functional Requirements

Your service architecture must expose a RESTful API with at least 2 endpoints. Services should communicate (where applicable) via HTTP, TCP or a simple message queue. A small front-end (or Postman test collection) should be provided to demonstrate service interaction (consider the structure of a set of specific User Acceptance Tests).

Non-Functional Requirements

Each service must run in its own Docker container.
Services must be defined in a docker-compose.yml file.
Include documentation and usage instructions in your repo.
System must support logging and basic error handling (optional).

Submission Requirements

A Zipped GitHub repo or local install with:

1. Code for all microservices
2. docker-compose.yml
3. Screenshots or screen recordings of the system running and any development processes or issues experienced.
4. README.md with instructions, directory structure and architectural diagram.

Suggested Directory Structure

```
<Your Student ID>-assignment2/
  .gitignore
  docker-compose.yml
  README.md (includes setup guide + architecture diagram)
  student-profile/
    app.py
    Dockerfile
    requirements.txt
  <name of your second service>/
    <your files>...
  <name of your third service>/
    <your files>...
  .env.example (if needed for assessors to copy and use)
```

Assessment Criteria

Criterion	Weight
Functional completeness (working APIs)	10

Assessment Criteria	Weight
Criterion	
Microservices architecture and separation	15
Dockerization and deployment	5
Documentation (README, diagrams, screen shots)	5
Code quality and modularity	5
Total	40

Bonus Ideas:

- Use environment variables for DB config via .env file.
- Use nginx or Traefik as a simple API gateway.
- Add basic authentication to one or more services.

Starter Files

To help you get started with this assignment you can download a partially developed microservices starter project (including **docker-compose.yml**, the first **student-profile** service **Flask api**. Access on GitHub here:

<https://github.com/JB-Connect/INFS605-Assignment2-Starter-Files>

Or on Canvas here:

https://canvas.aut.ac.nz/courses/20507/files/7107960/download?download_frd=1

This starter zip file includes docker-compose.yml, an initial student-profile Flask service with Dockerfile, and an example.env file. You can use this as a starting point for your assignment and consider adding further services – a Course Catalogue Service, a Feedback Service, a Notification Service, a Grades Service (that can store and retrieve grades for students and courses), a Timetable Service (that can show weekly schedules for students or lecturers) an Assignment Tracker (shows coming and submitted assignments), a Room Booking Service (for team meetings), or come more generic services such as – an Image Upload Service, a Search Service, a Logging or Audit Trail Service, Frontend UI, PDF Generator, System Metrics Service.

Please also note the following:

It is your responsibility to check assignments for grammar and spelling errors before submitting them. You do not want your lecturer's to be your assessment quality control process.

To improve the quality of your submissions, there is a great and free proofreading service **Studiosity** at AUT that you can use 24x7 to improve your writing - See the link on Canvas. Ideal for a proof read of your Readme.md file.

Late penalty

Late submissions will be accepted for up to five days after the due date. There will be a daily penalty of 5% applied to the final grade for each 24-hour period late.

Assessment Regulations

This paper operates according to the Faculty of Design and Creative Technologies guidelines and the Auckland University of Technology regulations as presented in the *General Academic Statute*. You are advised to become familiar with the contents of the AUT Academic Policies and Regulations via the General Academic Statute:

https://www.aut.ac.nz/data/assets/pdf_file/0004/119506/Calendar_2025.pdf

Academic Honesty and Integrity

The university regards most seriously any acts of dishonesty relating to assessments. Any such acts may result in a mark of zero for the assessment and may lead to disciplinary action. Examples of dishonesty include:

- Plagiarism.
- Unauthorised collaboration.
- Examination misconduct.
- Theft of another student's work.
- Use of previously submitted assignment work.

Plagiarism

Plagiarism means borrowing from the work of another without indicating by referencing (and by quotation marks where exact phrases are borrowed) that the ideas expressed are not one's own. Students may use the ideas and information of other authors, but this use must be acknowledged. It is not acceptable to submit an assignment which is simply paraphrasing extracts from other authors: the work submitted must include a recognisable intellectual contribution by the student.

Unauthorised Collaboration

Unauthorised collaboration means joint effort between students or students and others, in preparing material submitted for assessment, except where this has been approved by the course programme. Students are encouraged to discuss matters covered in classes, but when writing an assignment, report, essay, or other piece of assessed work, the recording and treatment of the data and the expression of ideas and central argument must be the student's own work.

Extensions and Special Consideration Applications (SCA)

Students may apply for an extension or for special consideration where there are exceptional circumstances. The SCA online request form is available from Canvas. You can access this via the "Special Consideration" tab on Canvas home once you have logged in. Follow the instructions provided.

Reconsideration

You are entitled to make a formal request for reconsideration of marks in cases where you believe your assessment work has not been marked in accordance with published marking criteria. This request can only be made when assessed work is returned.

Note that, where assessments have been submitted electronically, you normally have 14 days from the time the graded assessment is made available to you to apply for reconsideration.